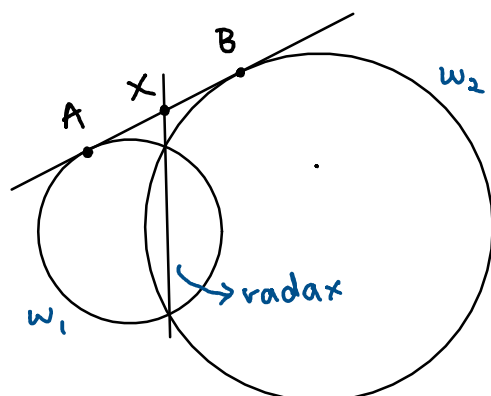


pop + radax problem-solving

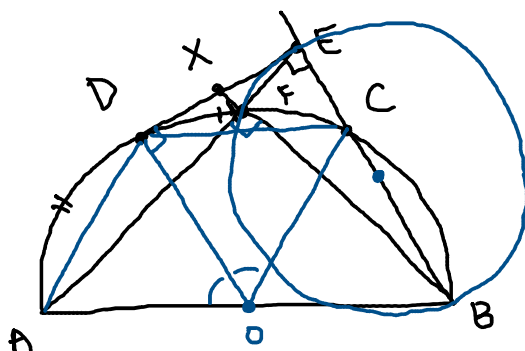
Sunday, 29 June, 2025 11:11 AM

1. Let Γ_1 and Γ_2 be two intersecting circles. Let a common tangent to Γ_1 and Γ_2 touch Γ_1 at A and Γ_2 at B . Show that the common chord of Γ_1 and Γ_2 , when extended, bisects segment AB .



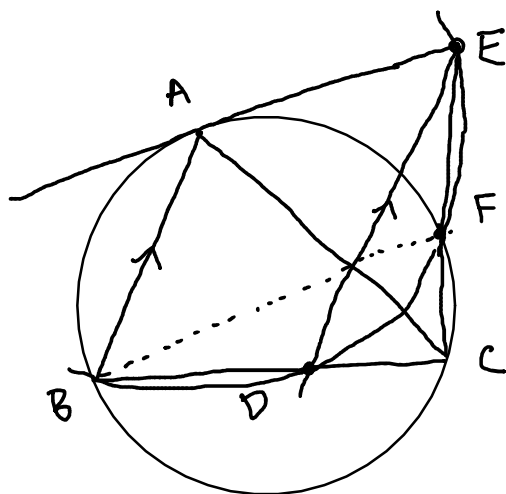
$$\begin{aligned}
 X &\in \text{radax} \\
 \Rightarrow \text{pow}(X, \omega_1) &= \text{pow}(X, \omega_2) \\
 \Rightarrow XA^2 &= XB^2 \\
 \Rightarrow XA &= XB \quad \blacksquare
 \end{aligned}$$

2. Let C be a point on a semicircle of diameter AB and let D be the midpoint of arc AC . Let E be the projection of D onto the line BC and F the intersection of line AE with the semicircle. Prove that BF bisects the line segment DE .



$$\begin{aligned}
 &DE \text{ tangent to } (BFE) \\
 \angle AOD &= \frac{\angle AOC}{2} = \frac{2\angle ABC}{2} = \angle ABC \\
 \Rightarrow DO &\parallel EB \Rightarrow \angle ODE = 90^\circ \\
 \Rightarrow DE &\text{ tangent to semi-circle} \\
 &\text{Done by Problem 1} \quad \blacksquare
 \end{aligned}$$

3. (APMO 2020 P1) Let Γ be the circumcircle of $\triangle ABC$. Let D be a point on the side BC . The tangent to Γ at A intersects the parallel line to BA through D at point E . The segment CE intersects Γ again at F . Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.

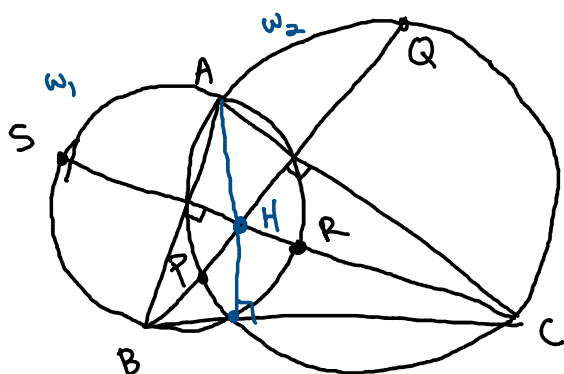


$$\angle EDC = \angle ABC = \angle EAC$$

$$\Rightarrow ADCE \text{ cyclic}$$

radax theorem on
 $(ABCF), (BDFE), (ADCE)$ ■

4. Let ABC be an acute triangle. Let the line through B perpendicular to AC meet the circle with diameter AC at points P and Q , and let the line through C perpendicular to AB meet the circle with diameter AB at points R and S . Prove that P, Q, R, S are concyclic.



$$AH = \text{radax of } \omega_1, \omega_2$$

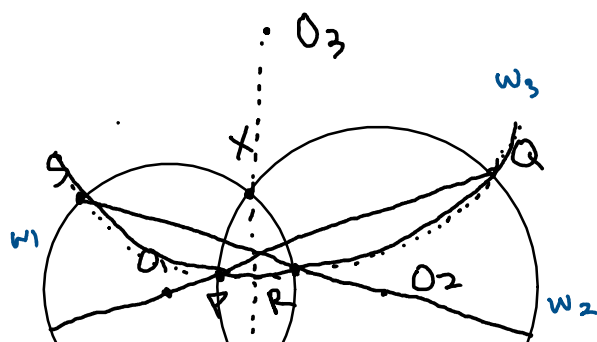
$$\text{pow}(H, \omega_1) = \text{pow}(H, \omega_2)$$

$$\Rightarrow -(HR \times HS) = -(HP \times HQ)$$

$$\Rightarrow HR \times HS = HP \times HQ$$

$$\therefore \text{by POP, } PQRS \text{ cyclic} \blacksquare$$

5. (USAMO 2009 P1) Given circles ω_1 and ω_2 intersecting at points X and Y , let ℓ_1 be a line through the center of ω_1 intersecting ω_2 at points P and Q and let ℓ_2 be a line through the center of ω_2 intersecting ω_1 at points R and S . Prove that if P, Q, R and S lie on a circle then the center of this circle lies on line XY .



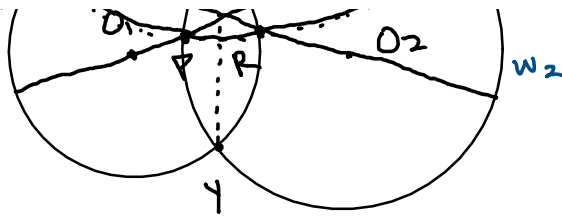
$$\text{goal: } \text{pow}(O_3, \omega_1) = \text{pow}(O_3, \omega_2)$$

$$\text{i.e. } O_3 O_1^2 - r_1^2 = O_3 O_2^2 - r_2^2$$

$$O_1 \in PQ = \text{radax of } \omega_2, \omega_3$$

$$\Rightarrow \text{pow}(O_1, \omega_2) = \text{pow}(O_1, \omega_3)$$

$$\Rightarrow O_1 O_2^2 - r_2^2 = O_1 O_3^2 - r_3^2$$



$$\Rightarrow \text{pow}(O_1, w_2) = \text{pow}(O_1, w_3)$$

$$\Rightarrow O_1 O_2^2 - r_2^2 = O_1 O_3^2 - r_3^2$$

$$O_2 \in RS = \text{radax of } w_1, w_3$$

$$\Rightarrow \text{pow}(O_2, w_1) = \text{pow}(O_2, w_3)$$

$$\Rightarrow O_2 O_1^2 - r_1^2 = O_2 O_3^2 - r_3^2$$

subtract:

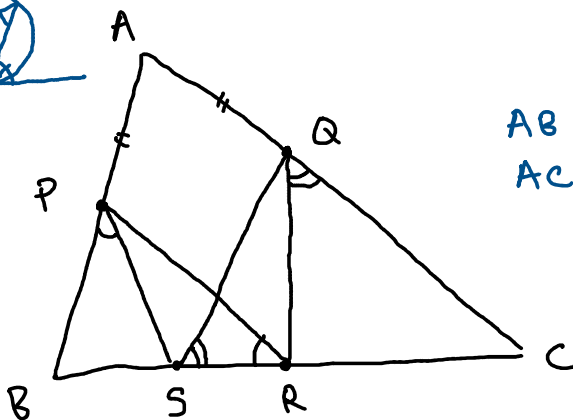
$$r_1^2 - r_2^2 = O_1 O_3^2 - O_2 O_3^2$$

$$\Rightarrow O_3 O_1^2 - r_1^2 = O_3 O_2^2 - r_2^2$$

$$\Rightarrow \text{pow}(O_3, w_1) = \text{pow}(O_3, w_2)$$

$$\therefore O_3 \in \text{radax } w_1, w_2 = XY \blacksquare$$

6. (USAJMO 2012 P1) Given a triangle ABC , let P and Q be points on segments $\overline{AB}$ and $\overline{AC}$, respectively, such that $AP = AQ$. Let S and R be distinct points on segment $\overline{BC}$ such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R, S are concyclic (in other words, these four points lie on a circle).



AB is tangent to (PSR)

AC is tangent to (QSR)

assume $PQRS$ not cyclic.

$RS = \text{radax of } (PSR), (QSR)$

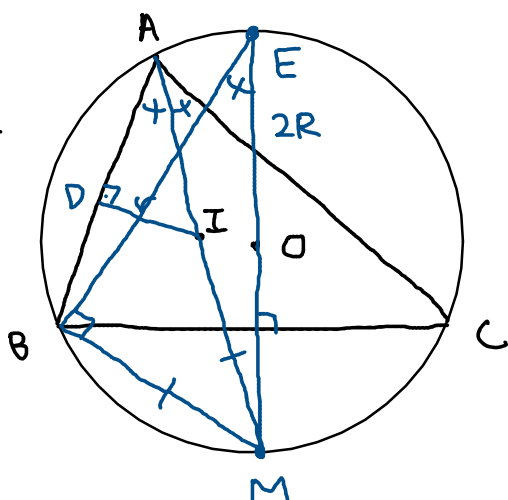
$$\text{pow}(A, (PSR)) = AP^2 = AQ^2 = \text{pow}(A, (QSR))$$

$$\Rightarrow A \in RS, \text{ contradiction}$$

$\therefore PQRS$ cyclic $\blacksquare$

7. (Euler's Theorem/Inequality) Let $\triangle ABC$ be a triangle. If O and I are its circumcenter and incenter, respectively, and R and r are its circumradius and inradius, respectively, then prove that $OI^2 = R(R - 2r)$.

circumradius and inradius, respectively, then prove that $OI^2 = R(R - 2r)$, and in particular $R \geq 2r$.



WTS:

$$OI^2 = R(R - 2r)$$

$$OI^2 = R^2 - 2Rr$$

$$\text{pow}(I, (ABC)) = OI^2 - R^2 = -2Rr$$

$$= -(AI \times IM)$$

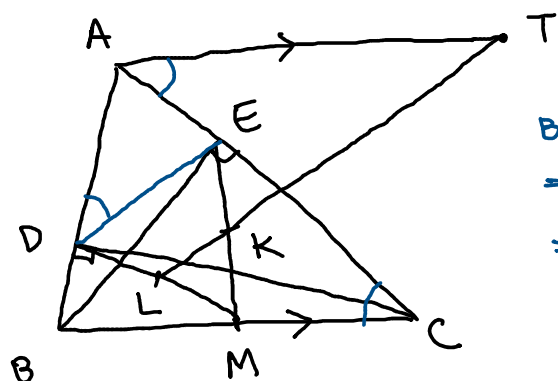
$$AI \times IM = 2Rr$$

By Incenter - Excenter Lemma,
 $MI = MB$

By AA criterion, $\triangle ADI \sim \triangle EBM$

$$\frac{BM}{DI} = \frac{ME}{AI} \Rightarrow \frac{IM}{r} = \frac{2R}{AI} \Rightarrow AI \times IM = 2Rr \quad \blacksquare$$

8. (Iran TST 2011 P1) In acute triangle ABC angle B is greater than C . Let M is midpoint of BC . D and E are the feet of the altitude from C and B respectively. K and L are midpoint of ME and MD respectively. If KL intersect the line through A parallel to BC in T , prove that $TA = TM$.



$BDEC$ cyclic

$$\Rightarrow \angle ADE = \angle C = \angle EAT$$

$\Rightarrow AT$ is tangent to (ADE)

Lemma: MD, ME tangent to (ADE)

Proof: Exercise $\Rightarrow$

$$KE = KM \Rightarrow KE^2 = KM^2$$

$$\Rightarrow \text{pow}(K, (ADE)) = \text{pow}(K, (M))$$

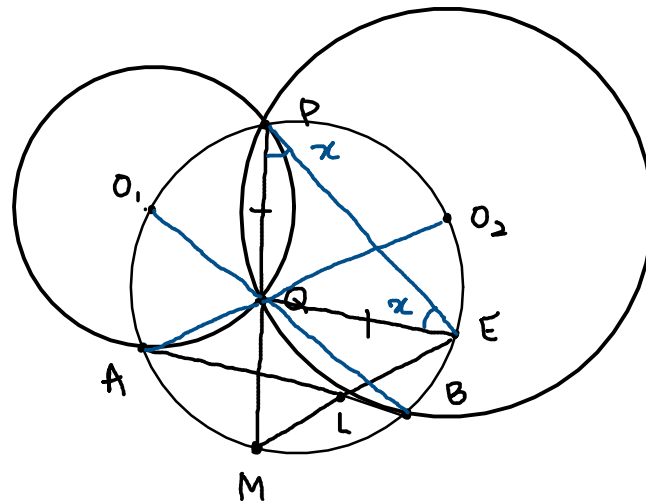
$$\text{Similarly, } \text{pow}(L, (ADE)) = \text{pow}(L, (M))$$

$$KL = \text{radax of } (ADE), (M)$$

$$T \in KL \Rightarrow \text{pow}(T, (ADE)) = \text{pow}(T, (M))$$

$$\Rightarrow TA^2 = TM^2 \Rightarrow TA = TM \quad \blacksquare$$

9. (Al-Khwarizmi 2024 P7) Two circles with centers O_1 and O_2 intersect at P and Q . Let ω be the circumcircle of the triangle PO_1O_2 ; the circle ω intersect the circles centered at O_1 and O_2 at points A and B , respectively. The point Q is inside triangle PAB and PQ intersects ω at M . The point E on ω is such that $PQ = QE$. Let ME and AB meet at L , prove that $\angle QLA = \angle MLA$.



$$\angle QAP = \frac{\angle QO_1P}{2} = \angle O_2O_1P = \angle O_2AP$$

$$\Rightarrow A-Q-O_2 \text{ collinear}$$

$$B-Q-O_1 \text{ collinear}$$

$$O_2P = O_2B \Rightarrow AQ \text{ bisects } \angle PAB$$

$$\text{similarly, } BQ \text{ bisects } \angle PBA$$

$$\therefore Q = \text{incenter } \triangle PAB$$

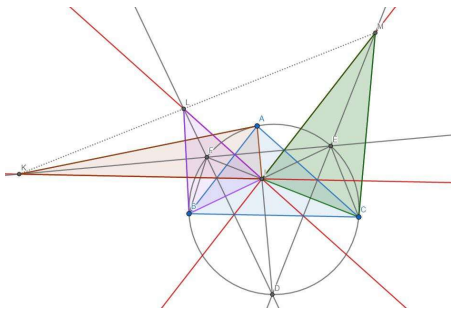
$$MQ = MA = MB \text{ (well-known)}$$

$$\begin{aligned} \angle MLA &= 180^\circ - \angle MAB - \angle AME \\ &= 180^\circ - \angle MAB - (180^\circ - \angle APE) \end{aligned}$$

$$\begin{aligned}
\angle MLA &= 180^\circ - \angle MAB - \angle AME \\
&= 180^\circ - \angle MAB - (180^\circ - \angle APE) \\
&= \angle APE - \angle MAB \\
&= \angle APE - \angle MPA = \angle MPE = x
\end{aligned}$$

By PS, $\angle QLA = x$
 $\Leftrightarrow \angle QLE = 180^\circ - 2x = \angle PQE$
 $\Leftrightarrow MP$ tangent to (QLE)
 $\Leftrightarrow MQ^2 = ML \times ME$
 $\Leftrightarrow MA^2 = ML \times ME$
 $\Leftrightarrow MA$ tangent to (ALE)
 $\Leftrightarrow \angle MAB = \angle MEA$
 $\Leftrightarrow MA = MB$,
 which is true. ■

10. (BMO 2015 P2) Let $\triangle ABC$ be a scalene triangle with incentre I and circumcircle ω . Lines AI, BI, CI intersect ω for the second time at points D, E, F , respectively. The parallel lines from I to the sides BC, AC, AB intersect EF, DF, DE at points K, L, M , respectively. Prove that the points K, L, M are collinear.



Guess: $K, L, M \in \text{radax of } (ABC), (I)$

$K \in \text{radax } (I), (ABC)$
 $\Leftrightarrow \text{pow}(K, (I)) = \text{pow}(K, (ABC))$
 $\Leftrightarrow KI^2 = KE \times KF$
 $\Leftrightarrow KI$ tangent to (IEF)
 $\Leftrightarrow \angle KIF = \angle FEI$
 $\Leftrightarrow \angle BCF = \angle BEF$,
 which is true.

Similarly, $L, M \in \text{radax of } (I), (ABC)$ ■